



ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 7: HealthConnect Testing, Refinement & End-to-End Validation

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 7
Project	HealthConnect Clinic Experience Lab
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Version	1.0
Issue Date	
Submission Deadline	End of Week 7 (Sunday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 7 Submission Form

1. BUSINESS SCENARIO

HealthConnect Clinic is a fictional healthcare provider seeking to reduce missed appointments, improve appointment attendance, make better use of appointment slots and provide more effective administrative support to patients.

During **Week 4**, interns established their understanding of the HealthConnect problem, reviewed the available project resources and defined their track-specific responsibilities.

During **Week 5**, interns moved from planning into practical implementation and produced initial project outputs across their respective professional tracks.

During **Week 6**, interns advanced their work through integration, advanced development and validation. They were required to connect their work with relevant tracks and demonstrate meaningful cross-track integration.

The Week 6 outputs now provide the foundation for the next stage of the HealthConnect project.

Week 7 moves the project into:

Testing - Refinement - End-to-End Validation

The focus is no longer simply on developing or integrating individual components.

Interns must now demonstrate that their work can be **tested systematically, refined based on evidence and validated as part of the wider HealthConnect solution.**

The overall project question remains:

How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

2. PROJECT CONTINUITY

From Week 4 onwards, all interns contribute to the same HealthConnect Experience Lab while working from the perspective of their respective professional tracks.

The project progresses as follows:

Week	Project Stage
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Week 4	Problem Understanding → Resource Review → Solution Planning
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Week 5	Analysis → Development → Initial Implementation
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Week 6	Integration → Advanced Development → Validation
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Week 7	Testing → Refinement → End-to-End Validation
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Week 8	Final Integration → Presentation
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Important

You must continue from your **Week 6 work.**

Do not:

- Start a new HealthConnect project.
- Select a new dataset.
- Discard your Week 6 work.
- Repeat your complete Week 5 or Week 6 analysis.
- Rebuild an identical Week 6 deliverable without meaningful improvement.
- Treat testing as simply describing what you built.

Your Week 7 work must demonstrate **forward progress from your Week 6 outputs**.

The purpose of Week 7 is to determine whether the work developed and integrated during previous weeks performs as intended, identify weaknesses, make appropriate improvements and validate the results.

3. WEEK 6 → WEEK 7 TRANSITION

Before beginning the Week 7 testing activities, briefly establish the connection between your Week 6 work and your Week 7 activities.

Identify:

1. Your main Week 6 output.
2. The most important component or result produced in Week 6.
3. The main issue, limitation or dependency identified in Week 6.
4. The component or workstream that requires testing.
5. The track(s) relevant to your Week 7 testing activities.
6. What you intend to test.
7. What you expect to improve or validate.

Do not reproduce your Week 6 report.

This should be a **concise transition into your Week 7 work**.

4. WEEK 7 ASSIGNMENT OBJECTIVE

The objective of Week 7 is to **systematically test, refine and validate the HealthConnect project components and their interaction within the wider multidisciplinary solution**.

By the end of Week 7, interns should be able to:

- Test an existing project component using appropriate methods.
- Identify errors, weaknesses, inconsistencies or gaps.
- Analyse testing results.
- Apply evidence-based improvements.
- Re-test improved outputs.
- Validate cross-track dependencies.
- Participate in multidisciplinary testing.
- Document testing results and decisions.
- Assess whether their component meets its intended purpose.
- Evaluate how their work contributes to the overall HealthConnect solution.
- Identify remaining issues before final integration and presentation.

5. WEEK 7 LEARNING OBJECTIVES

By completing this assignment, interns should be able to:

- Apply systematic testing methods to a real-world project.
- Evaluate technical, analytical or project outputs against defined requirements.
- Identify and investigate errors or unexpected results.
- Refine an existing solution based on evidence.
- Re-test outputs after making improvements.
- Work effectively within a multidisciplinary project team.
- Validate dependencies between different professional workstreams.
- Document testing evidence professionally.
- Make evidence-based decisions.
- Communicate issues and resolutions clearly.
- Assess solution readiness for final integration.
- Contribute meaningfully to an end-to-end project validation process.

6. PROJECT RESOURCES

Continue using the approved HealthConnect project resources:

6.1 HealthConnect Appointment Dataset

File: HealthConnect_Appointment_Data.csv

6.2 HealthConnect Data Dictionary

File: HealthConnect_Data_Dictionary.xlsx

6.3 HealthConnect Clinic Knowledge Base

File: HealthConnect_Clinic_Knowledge_Base.docx

These remain the approved HealthConnect project resources.

Do not overwrite the original project resources.

Any processed, transformed or derived files must remain separate and properly documented.

For Generative AI work, responses must remain within the approved HealthConnect Knowledge Base and must not invent clinic or medical information.

7. PART 1 — WEEK 7 TESTING READINESS

Before performing your track-specific testing, establish what you are going to test and why.

A. Identify the Component to Be Tested

Clearly identify:

- What Week 6 component are you testing?
- What was its intended purpose?
- Which requirement or objective should it satisfy?
- Which track(s) are connected to this component?

B. Define the Testing Objective

State:

What are you trying to determine through testing?

For example:

- Does the dashboard produce accurate KPI results?
- Does the model perform consistently on the test data?
- Does the ML pipeline correctly process inputs and produce the expected output?
- Does the AI assistant respond appropriately to supported and unsupported questions?
- Are project dependencies being completed as expected?

C. Define Expected Results

For each major test, identify:

1. Test/Scenario.
2. Expected result.
3. Actual result.
4. Whether the test passed or failed.
5. Issue identified, if any.
6. Action taken.
7. Retest result.

D. Establish Testing Evidence

Your testing should produce appropriate evidence, such as:

- Test results.
- Updated analysis.
- Updated dashboard.
- Model evaluation results.
- Error analysis.
- Code.

- Pipeline test output.
- AI assistant test cases.
- Before/after outputs.
- Screenshots.
- Updated project documentation.
- Issue log.
- GitHub commit where applicable.
- Validation record.

Simply stating that your work was tested is not sufficient.

8. PART 2 — TRACK-SPECIFIC ASSIGNMENTS

A. PROJECT MANAGEMENT TRACK

Week 7: Project Testing Coordination, Issue Management & Readiness

Week 6 focused on integrating the different HealthConnect workstreams, tracking dependencies and preparing for testing.

In Week 7, the Project Management track is responsible for **coordinating the testing and refinement stage** of the project.

Tasks

1. Review the Week 6 project status.
2. Review the Week 6 integration/dependency register.
3. Review the testing coordination plan prepared in Week 6.
4. Identify the HealthConnect components requiring testing.
5. Develop or update a Week 7 testing and validation plan.
6. Define testing activities across the project workstreams.
7. Identify dependencies between testing activities.
8. Coordinate testing activities between relevant tracks.
9. Establish and maintain an **Integration & Issue Log**.
10. Record issues identified during testing.
11. Assign responsibility for resolving relevant issues.
12. Track the status of identified issues.
13. Monitor unresolved cross-track dependencies.
14. Coordinate retesting after issues have been addressed.
15. Update project risks based on testing findings.
16. Update the project timeline based on actual progress.
17. Record major decisions made during testing and refinement.
18. Assess whether the project remains aligned with the HealthConnect objective.
19. Identify work that must be completed before Week 8.
20. Prepare a **Week 8 Project Readiness Assessment**.

Week 7 Output

HealthConnect Testing Coordination & Project Readiness Package

Include:

- Week 7 testing/validation plan.
- Updated project timeline.
- Testing activity tracker.
- Integration & issue log.
- Updated risk/dependency register.
- Issue resolution status.
- Decision log.
- Cross-track coordination record.
- Testing progress/status report.
- Week 8 readiness assessment.
- Outstanding actions before final integration.

B. DATA ANALYTICS TRACK

Week 7: Analytical Testing, KPI Validation & Dashboard Refinement

Week 5 established the initial EDA, KPIs, dashboard/visualisation and business insights.

Week 6 deepened the analysis, validated important findings and improved the dashboard.

Do not repeat the complete Week 5 EDA or simply recreate the Week 6 dashboard.

Week 7 should focus on **testing the accuracy, consistency and usefulness of the analytical outputs.**

Tasks

1. Review the Week 6 analytical outputs.
2. Identify the most important analytical outputs requiring testing.
3. Validate the calculations of your important KPIs.
4. Check whether dashboard values agree with the underlying data.
5. Test relevant dashboard filters and calculations.
6. Investigate unexpected or inconsistent analytical results.
7. Validate important findings against the underlying dataset.
8. Reassess the most important attendance/no-show findings.
9. Test whether important conclusions remain valid across relevant segments.
10. Identify any analytical errors or weaknesses.
11. Refine visualisations that are misleading, unclear or unnecessary.
12. Improve dashboard usability and business interpretation.
13. Reassess recommendations based on the testing results.
14. Identify which validated findings should influence the wider HealthConnect solution.
15. Share relevant validated findings with the appropriate HC-POD track(s).
16. Document any changes made as a result of testing.
17. Re-test important outputs after refinement.
18. Document remaining analytical limitations.

Week 7 Output

HealthConnect Analytics Testing & Refinement Report/Notebook

Include:

- Testing approach.
- KPI validation.
- Analytical validation results.
- Dashboard testing evidence.
- Refined dashboard/visualisations.
- Validated key findings.
- Updated business insights.
- Updated recommendations.
- Before/after improvements where applicable.
- Cross-track testing contribution.
- Limitations.
- Remaining analytical issues.
- Week 8 recommendations.

C. DATA SCIENCE TRACK

Week 7: Model Testing, Error Analysis & Refinement

Week 5 established the baseline classification model.

Week 6 focused on model improvement, error analysis, model comparison and validation.

Week 7 now moves into **systematic testing and refinement of the candidate model**.

Tasks

1. Review the Week 6 candidate model.
2. Review the model testing requirements identified in Week 6.
3. Define the appropriate testing approach.
4. Test the candidate model using appropriate test/validation data.
5. Evaluate appropriate classification metrics.
6. Analyse model errors.
7. Investigate false positives and false negatives where applicable.
8. Identify patterns in model errors.
9. Assess model performance across relevant segments where appropriate.
10. Review whether the selected features continue to support the prediction objective.
11. Identify any evidence of overfitting or other modelling weaknesses where appropriate.
12. Compare the candidate model against the Week 5 baseline.
13. Determine whether the observed improvement is meaningful.
14. Identify any remaining model weaknesses.
15. Incorporate relevant validated findings from Data Analytics where appropriate.
16. Communicate relevant model requirements/results to ML Engineering.
17. Make justified refinements where necessary.
18. Re-test the refined model.

19. Document the results before and after refinement.
20. Assess whether the model is suitable for the intended HealthConnect use case.
21. Document remaining model limitations and risks.
22. Define what must be completed before Week 8.

Week 7 Output

HealthConnect Model Testing & Refinement Notebook/Report

Include:

- Week 6 candidate model summary.
- Testing approach.
- Test results.
- Model performance metrics.
- Error analysis.
- False-positive/false-negative analysis where applicable.
- Baseline vs candidate/refined model comparison.
- Refinement activities.
- Retesting results.
- Interpretation.
- Business relevance.
- Model suitability assessment.
- Cross-track testing/integration evidence.
- Limitations and risks.
- Week 8 recommendations.

D. MACHINE LEARNING ENGINEERING TRACK

Week 7: ML Pipeline Testing, Reliability & Refinement

Week 6 focused on integrating the Data Science model into the ML pipeline and validating the interaction between preprocessing and model components.

Week 7 focuses on **broader pipeline testing, reliability and refinement**.

Tasks

1. Review the integrated ML pipeline from Week 6.
2. Review the Week 6 testing requirements.
3. Define appropriate pipeline test cases.
4. Test the data-processing workflow.
5. Test preprocessing components.
6. Test the interaction between preprocessing and the model.
7. Validate expected inputs.
8. Validate expected outputs.
9. Test appropriate invalid or unexpected input scenarios.
10. Identify pipeline failures or inconsistencies.
11. Investigate the cause of identified issues.
12. Resolve appropriate technical issues.

13. Improve error handling where necessary.
14. Improve logging or execution information where necessary.
15. Re-run tests after making changes.
16. Validate that the pipeline produces the expected output after refinement.
17. Check reproducibility using the documented setup.
18. Review configuration and dependency management.
19. Confirm compatibility with the Data Science model requirements.
20. Update repository documentation.
21. Record technical issues that remain unresolved.
22. Prepare the pipeline for final integration in Week 8.

Week 7 Output

HealthConnect ML Pipeline Testing & Reliability Package

Include:

- Integrated pipeline.
- Test plan/test cases.
- Test results.
- Input/output validation.
- Preprocessing validation.
- Model integration testing.
- Error-handling evidence.
- Issues identified.
- Fixes/refinements.
- Retest evidence.
- Updated repository.
- Configuration/dependency documentation.
- Technical documentation.
- Remaining technical issues.
- Week 8 readiness recommendations.

Important: Week 7 is not necessarily the final production deployment stage. The focus is **testing, reliability, refinement and readiness for final integration**.

E. GENERATIVE AI TRACK

Week 7: AI Assistant Testing, Safety Validation & Refinement

Week 5 established the initial AI assistant prototype and testing approach.

Week 6 evaluated and refined the assistant, including prompt improvements, additional scenarios, Knowledge Base compliance and escalation behaviour.

Week 7 now focuses on **systematic testing, safety validation, refinement and retesting**.

Tasks

1. Review the Week 6 AI assistant.

2. Review the testing criteria established in Week 6.
3. Develop or refine a structured AI assistant test set.
4. Test normal HealthConnect questions.
5. Test ambiguous questions.
6. Test unsupported requests.
7. Test requests requiring escalation.
8. Test situations where the Knowledge Base does not provide sufficient information.
9. Test consistency across similar questions.
10. Identify inaccurate, incomplete, unclear or inconsistent responses.
11. Identify weaknesses in the prompt/workflow.
12. Check compliance with the approved HealthConnect Knowledge Base.
13. Verify that the assistant does not invent clinic or medical information.
14. Refine prompts or workflow where necessary.
15. Re-test the assistant after making improvements.
16. Compare important before/after outputs.
17. Evaluate response consistency.
18. Evaluate escalation behaviour.
19. Evaluate how the assistant handles unsupported or unavailable information.
20. Document safety or reliability issues identified during testing.
21. Document improvements made.
22. Identify remaining limitations.
23. Coordinate relevant findings with the HC-POD.
24. Prepare the assistant for final HealthConnect integration.

Week 7 Output

HealthConnect AI Assistant Testing & Refinement Package

Include:

- AI assistant test set.
- Test scenarios.
- Expected vs actual responses.
- Evaluation results.
- Weak-response/error analysis.
- Safety testing evidence.
- Escalation testing evidence.
- Unsupported-request testing.
- Knowledge Base compliance evidence.
- Before/after examples.
- Prompt/workflow refinements.
- Retesting results.
- Remaining limitations.
- Cross-track testing contribution.
- Week 8 recommendations.

Important: The GenAI solution must remain within the approved HealthConnect Clinic Knowledge Base. Do not invent clinic or medical information.

9. MANDATORY HC-POD CROSS-TRACK TESTING & REFINEMENT

This is the major difference between **Week 6 and Week 7**.

Week 5

Identify the dependency

Week 6

Integrate the work

Week 7

Test and refine the integrated work

Each intern must complete **at least one meaningful cross-track testing or validation activity** within the HC-POD.

The purpose is to determine whether the different HealthConnect components work together as intended.

Each intern must document:

1. Track collaborated with.
2. Project dependency.
3. Component/output being tested.
4. Information/output received.
5. Information/output provided.
6. Testing activity completed.
7. Issue or finding identified.
8. Refinement/action taken.
9. Retest result.
10. What changed as a result.
11. Evidence.
12. How the activity improved the overall HealthConnect solution.

Important

Communication alone is not sufficient.

A meeting, chat, message or discussion does not automatically constitute meaningful collaboration.

The collaboration should result in a:

Test → Finding → Action → Retest → Validated Improvement

where applicable.

10. HEALTHCONNECT TESTING & VALIDATION MATRIX

Use the following as guidance:

Track	Potential Testing/Validation Connection
Data Analytics → Data Science	Validate analytical findings that influence model features or modelling decisions
Data Science → ML Engineering	Test whether the integrated pipeline correctly handles model requirements and expected inputs/outputs
ML Engineering → Data Science	Identify technical pipeline behaviour or integration issues that require model-side review
Data Analytics → Project Management	Report validated findings and issues that may influence project decisions
GenAI → Project Management	Report assistant testing results, safety issues, dependencies and required actions
Project Management → All Tracks	Coordinate testing activities, issues, dependencies, timelines and retesting
All Tracks → All Tracks	Conduct relevant end-to-end validation where one component depends on another

These are examples, not instructions that every intern must follow exactly.

Your actual collaboration should be based on your **Week 6 work and the dependencies within your HC-POD**.

11. END-TO-END HEALTHCONNECT VALIDATION

The purpose of end-to-end validation is to determine whether the individual HealthConnect components contribute coherently to the overall project objective.

The HC-POD should consider:

Data Analytics

- Are the important findings accurate?
- Are the KPIs correctly calculated?
- Are the insights useful for decision-making?
- Does the dashboard communicate the findings clearly?

Data Science

- Has the model been appropriately tested?
- Are the model's strengths and weaknesses understood?
- Is the model suitable for its intended HealthConnect use case?

ML Engineering

- Does the pipeline correctly process the required inputs?
- Does the model integrate correctly?
- Are outputs reliable and reproducible?

Generative AI

- Does the assistant respond appropriately?
- Does it remain within the approved Knowledge Base?
- Does it handle unsupported and escalation scenarios safely?

Project Management

- Are issues being tracked?
- Are dependencies being resolved?
- Are responsibilities clear?
- Is the project ready for final integration?

End-to-End Validation Questions

Each intern should contribute to answering:

- 1. What component did I test?**
- 2. What did I expect to happen?**
- 3. What actually happened?**
- 4. What issue or weakness did I identify?**
- 5. What did I change or improve?**
- 6. What happened when I tested it again?**
- 7. Which other track was involved?**
- 8. How does the validated component contribute to HealthConnect?**
- 9. What still needs to be completed before Week 8?**

These responses must be supported by appropriate evidence.

12. TESTING, REFINEMENT & VALIDATION RECORD

Each intern should maintain a testing record.

Field	Description
Component Tested	What was tested
Testing Objective	What the test was intended to determine
Test/Scenario	What was tested
Expected Result	What should happen
Actual Result	What happened
Pass/Fail	Whether the test met the expected result
Issue Identified	Problem or weakness discovered
Action Taken	Improvement or correction made
Retest Result	Result after refinement
Collaborating Track	Relevant HC-POD track
Evidence	Screenshot, code, notebook, dashboard, model output, test record, GitHub commit, etc.

Where no issue is identified, clearly document the successful validation rather than inventing an issue.

13. ASSUMPTIONS, LIMITATIONS, RISKS & DEPENDENCIES

Review the issues identified during Weeks 4, 5 and 6.

Document:

- Issues that have been resolved.
- Issues that remain unresolved.
- New issues discovered during testing.
- New cross-track dependencies.
- Technical limitations.
- Data limitations.
- Analytical limitations.
- Modelling limitations.
- AI safety/reliability concerns where applicable.

- Integration challenges.
- Testing limitations.
- Potential impact on the project.
- Recommended mitigation.
- Issues that must be addressed before Week 8.

Do not create a separate report if these items can be incorporated into your main Week 7 deliverable.

14. WEEK 7 PROJECT SUMMARY

Prepare a concise **Week 7 Project Summary** covering:

1. What you planned to test.
2. What you actually tested.
3. The most important testing results.
4. Issues or weaknesses identified.
5. Improvements/refinements made.
6. Retesting results.
7. Which track(s) you collaborated with.
8. What was tested collaboratively.
9. What changed as a result.
10. Key findings or validation outcomes.
11. Major challenges encountered.
12. Important decisions made.
13. Remaining limitations.
14. Remaining issues or dependencies.
15. Your contribution to the overall HealthConnect project.
16. What must be completed before Week 8.

The summary should complement your main deliverable and should **not repeat it**.

15. SUBMISSION REQUIREMENTS

Your Week 7 submission must contain:

1. Track-Specific Week 7 Output

Submit the required deliverable for your assigned track.

2. Week 7 Project Summary

Submit the concise project summary described above.

3. Testing & Validation Evidence

Provide evidence supporting your testing and refinement activities.

4. HC-POD Cross-Track Evidence

Provide evidence of at least one meaningful cross-track testing, refinement or validation activity.

5. Supporting Files

Where applicable:

- Reports
- Notebooks
- Dashboards
- Code
- Models
- Pipelines
- Prototypes
- Test results
- Screenshots
- Project management artefacts
- Issue logs
- Documentation

6. Google Drive

Upload the required files to your designated project folder.

Ensure:

- Correct files are uploaded.
- Files are properly named.
- Folders are organised.
- Mentor access is available where applicable.

7. GitHub

Where applicable:

- Commit your Week 7 work.
- Update `README.md`.
- Maintain an organised repository.
- Ensure relevant files are accessible.
- Clearly document important changes.

16. PROFESSIONAL DEVELOPMENT REQUIREMENT

Continue maintaining your HealthConnect project professionally.

Interns should:

- Maintain organised project folders/repositories.
- Document important decisions.
- Maintain appropriate version control.
- Clearly document changes between weeks.
- Update README .md where applicable.
- Maintain appropriate naming conventions.
- Present testing results professionally.
- Clearly distinguish between previous and updated versions.
- Share meaningful project progress on LinkedIn/X where appropriate.
- Accurately represent their actual contribution to the project.

17. Assessment Criteria

Criteria	Weight
Application of Previous Project Work	10%
Track-Specific Technical/Analytical Quality	25%
Testing & Validation Quality	20%
Refinement & Problem-Solving	15%
HC-POD Cross-Track Collaboration	10%
Documentation & Evidence	10%
Professional Presentation	10%
Total	100%

18. IMPORTANT WEEK 7 REMINDER

WEEK 7 IS NOT A REPEAT OF WEEK 6.

You are expected to **build forward, not start again.**

Your progression should be:

Week 6 Output → Test → Identify Issues → Refine → Re-test → Validate → Document → Prepare for Week 8

You are **not expected to complete the entire HealthConnect solution this week.**

The purpose of Week 7 is to demonstrate that your individual contribution and its connections with other workstreams have been **tested, refined and validated sufficiently to move toward final integration.**

Your work should show:

Individual Expertise + Cross-Track Testing + Evidence-Based Refinement + End-to-End Validation

19. KEY DIFFERENCE BETWEEN WEEK 6 AND WEEK 7

Week 6	Week 7
Integration	Testing
Advanced development	Refinement based on testing
Validate integration	Validate end-to-end behaviour
Improve components	Test and improve components
Identify integration issues	Identify functional/quality issues
Resolve integration issues	Refine and re-test
Actually, integrate with another track	Test/refine with another track
Prepare for testing	Demonstrate testing evidence
Prepare for Week 7	Prepare for Week 8 final integration

The Week 6 assignment already establishes this progression and specifically identifies Week 7 as the **testing stage following integration and validation**.

Professional Development Requirement

As part of your professional development, you should:

- Update your existing GitHub repository.
- Update your `README.md`.
- Document your analytical process.
- Present your dashboard professionally.
- Share your learning and project progress on LinkedIn and X where appropriate.
- Tag AnalystLab Africa where applicable.
- Use **#AnalystLabAfrica** where appropriate.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.